

■ 1.8 Graphics

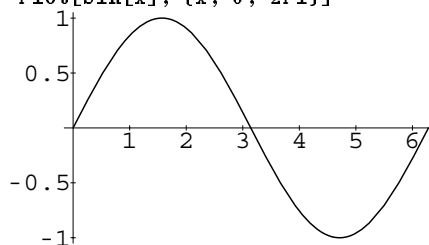
■ 1.8.1 Basic Plotting

`Plot[f, {x, xmin, xmax}]` plot f as a function of x from $xmin$ to $xmax$

`Plot[{f1, f2, ...}, {x, xmin, xmax}]`
plot several functions together

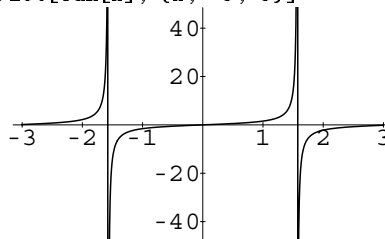
Basic plotting functions.

This plots a graph of $\sin(x)$ as a function of x from 0 to 2π . `In[1]:= Plot[Sin[x], {x, 0, 2Pi}]`



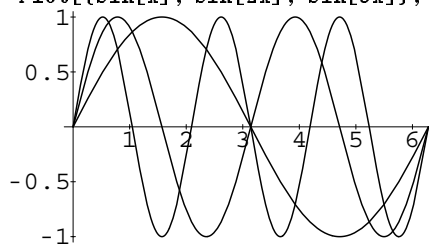
You can plot functions that have singularities. *Mathematica* will try to choose appropriate scales.

`In[2]:= Plot[Tan[x], {x, -3, 3}]`



You can give a list of functions to plot.

`In[3]:= Plot[{Sin[x], Sin[2x], Sin[3x]}, {x, 0, 2Pi}]`



On many systems, you have to “initialize” graphics in *Mathematica*, specifying for example, what kind of graphics device you are using. The information which came with your copy of *Mathematica* should tell you how to do what is needed.

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; <http://www.wolfram.com/>; 1-800-441-6284.

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Section ?? discusses some general features of graphics initialization in *Mathematica*.